MCV4U – Course Culminating Task – Guidelines

Task: You will create a website using any web building site/app to teach and assess a broad topic learned in calculus this semester. Your web page must include:

- Teacher profiles (you and your partner). This part is optional.
- At least two Instructional video lessons. Each video should be at least 8 minutes in length, or longer. The **video must have visual appeals. I will show a sample video in class**
- Sufficient amount of other instructional materials such as text, graphs, images, PowerPoint, links to external resources, etc.
- Sufficient amount of homework (minimum 4 homework) for all the topics you cover in your website, with step-by-step solutions provided. This is to provide practice opportunity for visitors to your website.
- At least 3 assessments, with solutions (steps shown). The assessments could include quizzes, games, or other methods of assessments.
- All assessments must include one or more of all 4 multiple categories of learning and teaching (KATC).
- One final assessment (similar to a final exam or a unit test) that includes all four categories of teaching and learning (KATC) with steps and solutions.
- Solutions to all homework, assessments and final exam must be typed. Hand written work is not acceptable.
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Partner:

Projects must be completed in groups of two. Find a partner from your class. You will prepare a short presentation about your project to the entire class. It is important that you and your partner have the same schedule.

Assessment Rubric.

Read and follow the rubric carefully (separate document). Both partners must contribute equally to all features of the website, and are responsible for understanding educational and technical content of the website.

Deadline #1: Friday October 16.

All potential issues such as problems related to working with your partner, technical difficulties in creating the website, clarification regarding the content and quality of your website, changing the topic , etc. must be discussed with me by **Friday October 16.** This is to allow enough time for me and you to consider alternatives. After the deadline I am unable to accommodate any changes related to your project.

Deadline #2: Sunday November 15 (11:59 pm).

Submit a link to your website and a pdf version of your website, before the final due date: **Sunday November 15 (11:59 pm).**

Suggested list of topics:

Choose one from the list of topics given below.

- No topics may be repeated by the groups that are in the same class, unless there are more groups than topics. Topics are assigned by First-come/first-serve.
- The topics and subtopics are general in nature. You need to research and examine them closely (consult textbook first) and understand what details need to be included on your website.

A- Rates of Change • A.R.O.C • I.R.O.C • Connection, relation, challenges • Historical perspective in finding the I.R.O.C • Limits and historical introduction • Existence of limits • Properties of limits • Continuity • Piecewise functions • Examples, applications	**B- Derivatives** • Existence of limit • Continuity • Differentiability • Connections, relations • Derivative rules, all of them (power rule, quotient rule, product rule, chain rule etc.) • Three applications of derivatives to real life
C- Derivatives and Optimization • Differentiability • Velocity and acceleration, accelerating, decelerating moving bodies • Maxima, Minima, Extreme Values • Optimization examples in **design, economics, and biological science**. Must include example from each field • Applications	**D- Curve Sketching** • Increasing/ Decreasing Functions • Critical Points • Local and Absolute Maxima and Minima • Vertical, Horizontal, Oblique Asymptotes • Concavity • Putting it all together • Applications
E- Derivative of Exponential, Logarithmic, and Trigonometric Functions • Differentiability • Derivative of exponential functions • Derivative of sinusoidal functions • Number e and the Derivative with Base **e** • Change in base in exponential functions • Derivative of tan and cot • Derivative of logarithmic functions • Applications	**F- Vectors** • What are vectors • Vector addition and subtraction in two and three space • Properties of vectors • Scalar Multiplication. Dot and Cross product of vectors in two and three space • Algebraic and Geometric vector in two and three space • Linear Combination and Spanning • Application Tension • Parallel, collinear, coplanar, scalar multiple , resultant, equilibrant vectors etc

G- Application of vectors • What are vectors • Geometric and algebraic vectors • Vector addition and subtraction in two and three space • Scalar Multiplication. Dot and Cross product of vectors in two and three space • All applications of Dot Product • All applications of Cross Product • Application Navigation (boat and airplane)	**H- Lines and Planes** • Vector, parametric, cartesian, and symmetric equations of a line in R2 and R3 • Vector, parametric and Cartesian, equations of a plane • Sketching planes and plotting points in R3 • Different ways to define a plane • Intersections of lines and planes in R3 • Intersections of two planes in R3. • Distance from a point to a line • Distance between a point and a line • Examples and Problems • Consistent and inconsistent systems
I- Intersection of Lines, Two Planes and Three Planes • Intersection of lines in R2 and R3 • Intersection of lines and planes in R3 • Systems of equations (Algebra Review) • Consistent and inconsistent systems • Intersection of two planes • Intersection of three planes • Parallelepiped • Distance between two planes • Distance between two skew lines	